



## **ARTIFICIAL-INTELLIGENCE-SUMMER-OF-CODE-NAVTTTC [Course]**

**Week 9 – Machine Learning –**

- ☐ **Ensemble Learning & its Types**
- ☐ **Random Forest Algorithm**

**[See examples / code in GitHub code repository]**

**It is not about Theory, it is 20% Theory and 80% Practical –  
Technical/Development/Programming [Mostly Python based]**

# ML – Ensemble Learning

Ensemble learning is a method where we use many small models instead of just one. Each of these models may not be very strong on its own, but when we put their results together, we get a better and more accurate answer. It's like asking a group of people for advice instead of just one person—each one might be a little wrong, but together, they usually give a better answer.

## Types of Ensembles Learning in Machine Learning

### Bagging (Bootstrap Aggregating):

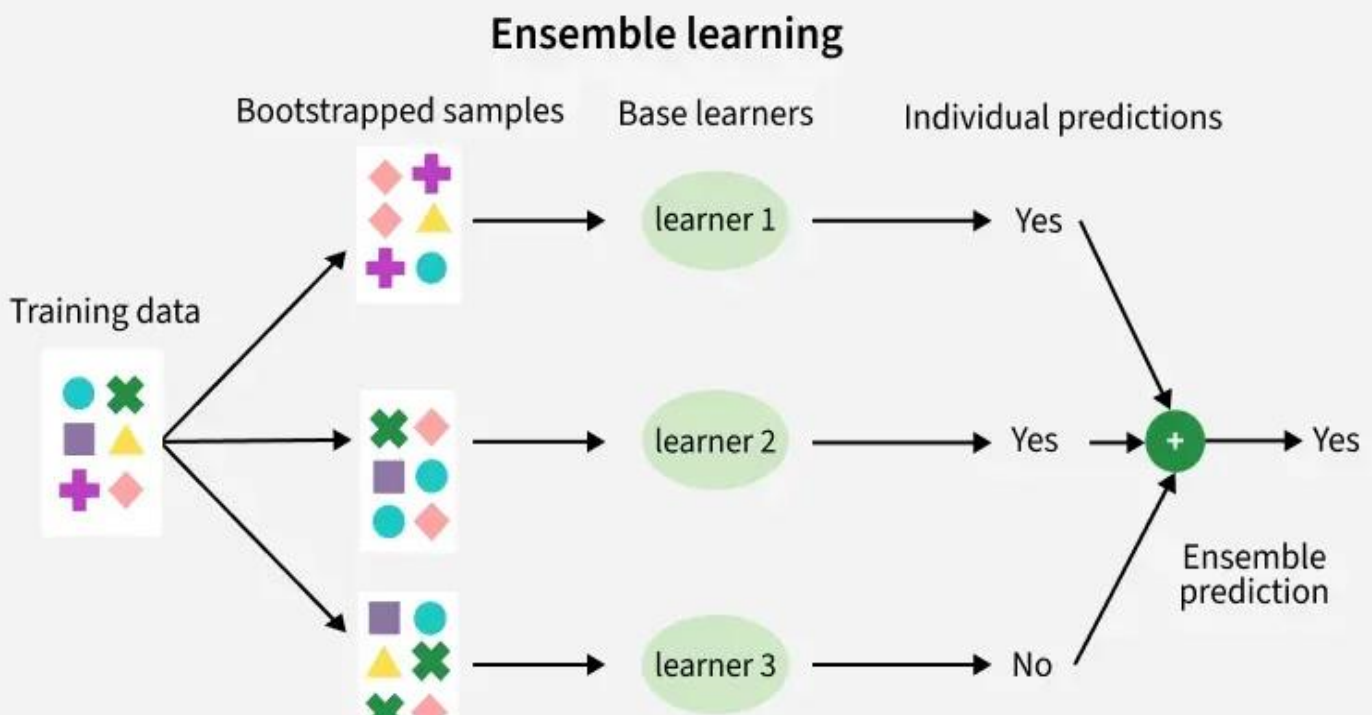
Models are trained independently on different random subsets of the training data. Their results are then combined—usually by averaging (for regression) or voting (for classification). This helps reduce variance and prevents overfitting.

### Boosting:

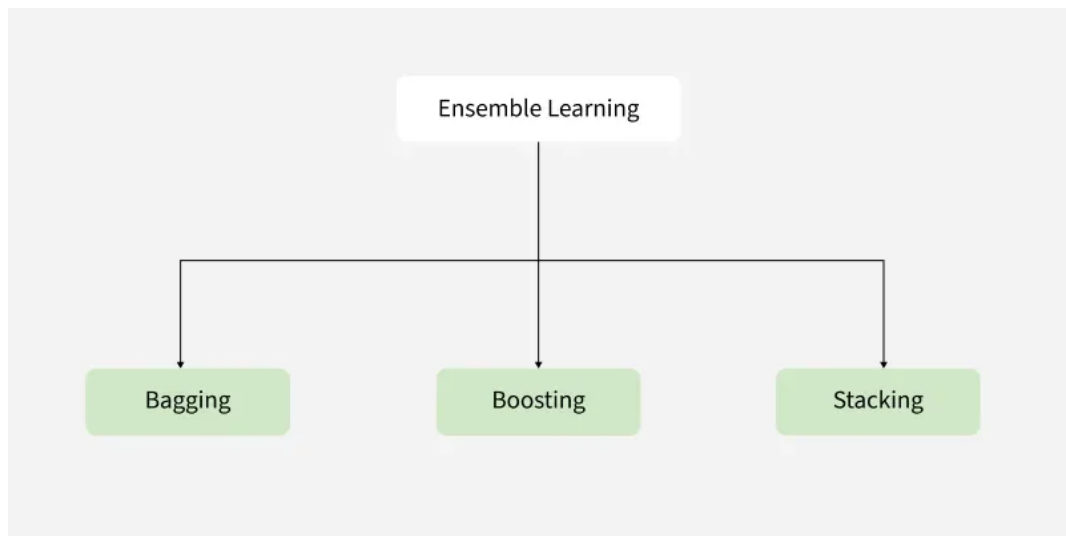
Models are trained one after another. Each new model focuses on fixing the errors made by the previous ones. The final prediction is a weighted combination of all models, which helps reduce bias and improve accuracy.

### Stacking (Stacked Generalization):

Multiple different models (often of different types) are trained, and their predictions are used as inputs to a final model, called a meta-model. The meta-model learns how to best combine the predictions of the base models, aiming for better performance than any individual model.



# ML – Ensemble Learning



## Bagging

Different Subset  
of the Training Data

Base Models

Most Frequent or  
Average Prediction

Prediction



## Boosting

Base Models

Prediction

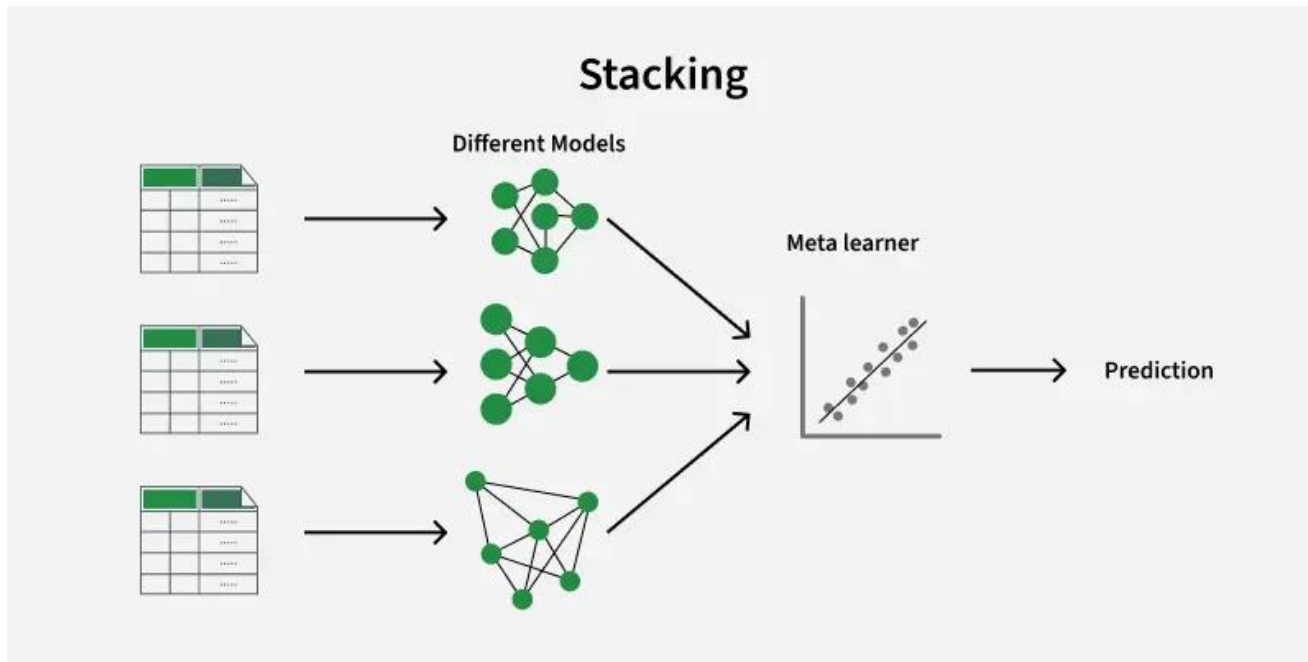
Prediction

Prediction



The training is modified  
based on the predictions

# ML – Ensemble Learning



## Benefits of Ensemble Learning in Machine Learning

Ensemble learning is a versatile approach that can be applied to machine learning model for: -

- **Reduction in Overfitting:** By aggregating predictions of multiple model's ensembles can reduce overfitting that individual complex models might exhibit.
- **Improved Generalization:** It generalizes better to unseen data by minimizing variance and bias.
- **Increased Accuracy:** Combining multiple models gives higher predictive accuracy.
- **Robustness to Noise:** It mitigates the effect of noisy or incorrect data points by averaging out predictions from diverse models.
- **Flexibility:** It can work with diverse models including decision trees, neural networks and support vector machines making them highly adaptable.
- **Bias-Variance Tradeoff:** Techniques like bagging reduce variance, while boosting reduces bias leading to better overall performance.



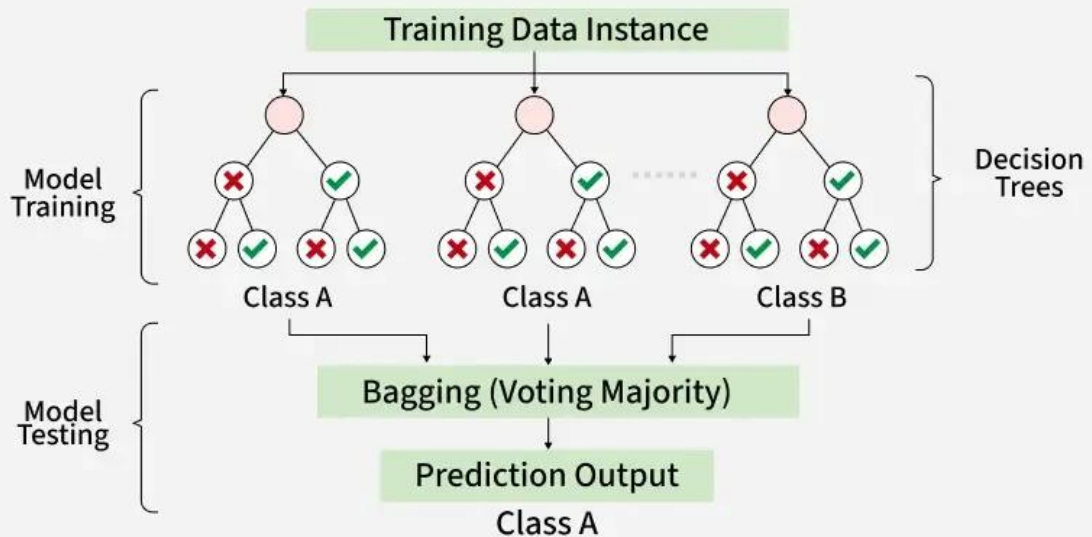
# ML – Random Forest for Classification

**Random Forest** is a machine learning algorithm that uses many decision trees to make better predictions. Each tree looks at different random parts of the data and their results are combined by voting for classification or averaging for regression. This helps in improving accuracy and reducing errors.

## Implementing **Random Forest for Classification Tasks**

Here we will predict survival rate of a person in titanic.

- ❑ Import libraries and load the Titanic dataset.
- ❑ Remove rows with missing target values ('Survived').
- ❑ Select features like class, sex, age, etc and convert 'Sex' to numbers.
- ❑ Fill missing age values with the median.
- ❑ Split the data into training and testing sets, then train a Random Forest model.
- ❑ Predict on test data, check accuracy and print a sample prediction result.



# ML – Random Forest for Regression

## Implementing Random Forest for Regression Tasks

- ❑ We will do house price prediction here.
- ❑ Load the California housing dataset and create a DataFrame with features and target.
- ❑ Separate the features and the target variable.
- ❑ Split the data into training and testing sets (80% train, 20% test).
- ❑ Initialize and train a Random Forest Regressor using the training data.
- ❑ Predict house values on test data and evaluate using MSE and  $R^2$  score.
- ❑ Print a sample prediction and compare it with the actual value.

## Advantages of Random Forest

- ❑ Random Forest provides very accurate predictions even with large datasets.
- ❑ Random Forest can handle missing data well without compromising with accuracy.
- ❑ It doesn't require normalization or standardization on dataset.
- ❑ When we combine multiple decision trees it reduces the risk of overfitting of the model.

## Limitations of Random Forest

- ❑ It can be computationally expensive especially with a large number of trees.
- ❑ It's harder to interpret the model compared to simpler models like decision trees.

<https://www.geeksforgeeks.org/machine-learning/random-forest-algorithm-in-machine-learning/>  
<https://www.ibm.com/think/topics/random-forest>



python

# ML | Difference Between Boosting Algorithms

| Algorithms                     | Gradient Boosting                               | CatBoost  |           |                                             |           |
|--------------------------------|-------------------------------------------------|-----------|-----------|---------------------------------------------|-----------|
|                                |                                                 | AdaBoost  | XGBoost   |                                             | LightGBM  |
| Year                           | -                                               | 1995      | 2014      | 2017                                        | 2017      |
| Handling Categorical Variables | May require preprocessing like one-hot encoding | No        | NO        | Automatically handles categorical variables | No        |
| Speed/Scalability              | Moderate                                        | Fast      | Fast      | Moderate                                    | Fast      |
| Memory Usage                   | Moderate                                        | Low       | Moderate  | High                                        | Low       |
| Regularization                 | NO                                              | No        | Yes       | Yes                                         | Yes       |
| Parallel Processing            | No                                              | No        | Yes       | Yes                                         | Yes       |
| GPU Support                    | No                                              | No        | Yes       | Yes                                         | Yes       |
| Feature Importance             | Available                                       | Available | Available | Available                                   | Available |

## Reference:

<https://www.geeksforgeeks.org/machine-learning/gradientboosting-vs-adaboost-vs-xgboost-vs-catboost-vs-lightgbm/>

<https://www.analyticsvidhya.com/blog/2020/02/4-boosting-algorithms-machine-learning/>

<https://www.linkedin.com/advice/0/how-do-you-compare-contrast-different-boosting>



# ML | Boosting vs Bagging

Bagging is another ensemble learning technique used to improve accuracy. Unlike bagging where models are trained independently on different machine learning models, Boosting models are trained sequentially with each model correcting the errors of its predecessor

| Feature                | Boosting                                                  | Bagging                                              |
|------------------------|-----------------------------------------------------------|------------------------------------------------------|
| Combination Type       | Combines predictions of different weak models             | Combines predictions of the same type of model       |
| Goal                   | Reduces <b>bias</b>                                       | Reduces <b>variance</b>                              |
| Model Dependency       | New models depend on previous models' errors              | All the models have the same weightage               |
| Weighting              | Models are weighted based on performance                  | All models have equal weight.                        |
| Training Data Sampling | Each new model focuses more on the misclassified examples | Each model is trained on random subset of data       |
| Error handling         | Focuses on correcting errors made by previous models      | Averages out errors from multiple models             |
| Parallelism            | Models are built sequentially less parallelizable         | Models can be built in parallel                      |
| Overfitting            | Less prone to overfitting with proper regularization      | Can be prone to overfitting with complex base models |
| Model Complexity       | Typically uses simpler models (like decision stumps)      | Can use complex models (like full decision trees)    |
| Example                | AdaBoost, Gradient Boosting, XGBoost, LightGBM            | Random Forest, Bagged Decision Trees                 |

**Reference:**

<https://www.geeksforgeeks.org/machine-learning/What-is-Bagging-classifier/>



# Exercises

See code here: <https://github.com/ShahzadSarwar10/ARTIFICIAL-INTELLIGENCE-SUMMER-OF-CODE-NAVTTTC-Sept2025/tree/main/Week9>

You should be able to analyze – each code statement, you should be able to see trace information – at each step of debugging. “DEBUGGING IS BEST STRATEGY TO LEARN A LANGUAGE.” So debug code files, line by line, analyze the values of variable – changing at each code statement. BEST STRATEGY TO LEARN DEEP.

Let's put best efforts.

Thanks.

Shahzad – Your AI – ML Instructor

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## Exercises



# python



Thank you - for listening and participating

☐ Questions / Queries

☐ Suggestions/Recommendation

☐ Ideas.....?

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